

Question:

A 7-year-old boy presents to your office with facial eczema. He has a history of recurrent infections, including multiple episodes of pneumonia that lasted several weeks and otitis media. Laboratory measurements of serum immunoglobulins show increased IgE and IgA but decreased IgM. Which of the following additional abnormalities would you expect to observe in this patient?

Answer Choices:

- A. Thrombocytopenia
- B. Leukopenia
- C. Anemia
- D. NADPH oxidase deficiency

Reasoning + Answer:

mobx

The patient is a 7-year-old boy with facial eczema, recurrent pneumonia, and otitis media, along with laboratory findings of increased IgE and IgA and decreased IgM. This clinical picture is highly suggestive of a selective immunodeficiency, specifically common variable immunodeficiency (CVID), though the pattern of immunoglobulin abnormalities (elevated IgE and IgA, decreased IgM) is more characteristic of a hyper-IgE syndrome, particularly in the context of recurrent infections and eczema.

Hyper-IgE syndrome is a primary immunodeficiency characterized by:

- Markedly elevated IgE
- Normal or low IgG and IgA (though IgA may be normal or increased)
- Decreased IgM
- Recurrent skin infections (eczema), sinopulmonary infections (pneumonia, otitis), and often a distinctive facial appearance

The hallmark of hyper-IgE syndrome is a defect in the IL-1 receptor signaling pathway, which leads to impaired Th17 cell function and defective antifungal and antibacterial defense. This condition is often associated with mutations in the STAT3 gene, which is critical for signaling downstream of cytokine receptors, including IL-10 and IL-6.

In this context, we are asked which additional abnormality we would expect. Among the choices:

- Thrombocytopenia (A): not typical in hyper-IgE syndrome
- Leukopenia (B): not a consistent feature; leukocytosis may occur during infections
- Anemia (C): not a hallmark
- NADPH oxidase deficiency (D): this causes chronic granulomatous disease (CGD), which presents with recurrent bacterial/fungal infections and is associated with low IgG and IgA, not elevated IgE. CGD does not have the IgE elevation seen here.

Thus, the IgE elevation and recurrent infections point strongly toward hyper-IgE syndrome, which is associated with STAT3 mutations, not NADPH oxidase deficiency.

Therefore, none of the listed abnormalities are directly caused by NADPH oxidase deficiency. But among the choices, the correct answer must reflect a known association.

However, in hyper-IgE syndrome, patients may present with leukocytosis during infection, but not leukopenia. The key is that NADPH oxidase deficiency is not present in this patient. This is a common distractor.

Given the question is asking what we would expect to observe and the answer choices are listed, the correct answer is not clearly one of the others, but the question may be testing if you recognize that hyper-IgE syndrome is not associated with NADPH oxidase deficiency. The correct answer is thus not D.

But the patient has elevated IgE and decreased IgM, which is classic for hyper-IgE syndrome. and it is not associated with NADPH oxidase deficiency.

Therefore, the correct answer is not D.

Wait. the question is asking what would we expect to observe? The answer must be one of the listed.

But the only abnormality that is commonly associated with hyper-IgE syndrome is no leukopenia. actually, it's often leukocytosis.

But the question asks: "Which would you expect to observe?"

The correct answer is not A, B, or C. because these are not typical.

But the key is: NADPH oxidase deficiency is associated with CGD. not hyper-IgE syndrome. So D is incorrect.

In hyper-IgE syndrome, patients often have normal or elevated white cells, so leukopenia is not expected.

Therefore, the best answer is that none of the above is correct. but only D is incorrect.

But the question is: which would you expect to observe?

In fact, patients with hyper-IgE syndrome may have mild leukopenia, but it's